AnyAttack: Towards Large-scale Self-supervised Adversarial Attacks on Vision-language Models

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Abstract

Vision-language models (VLMs) enable powerful multimodal capabilities but remain vulnerable to targeted adversarial images. Existing targeted attacks typically require labels or text guidance, limiting scalability and real-world relevance. We introduce AnyAttack, a transfer-based black-box attack that reframes targeted attacks as a foundation modeling problem. AnyAttack pre-trains a universal adversarial noise generator in a self-supervised manner by aligning noised random carriers to target image embeddings, removing the need for target labels or text. We pre-train on LAION-400M and fine-tune with frozen auxiliary encoders to enhance cross-model transfer. In a single forward pass, AnyAttack produces perturbations bounded by ℓ_∞ with $\epsilon = 16/255$ that induce desired targets on diverse VLMs. On MSCOCO image–text retrieval, AnyAttack-Bi with auxiliary encoders achieves R@Mean 35.88 on CLIP ViT-B/16, 33.66 on ViT-L/14, and 33.18 on ViT-L/14×336, outperforming baselines by large margins. On SNLI-VE multimodal classification it attains 44.8% accuracy versus 24.8% (best baseline), and it improves captioning metrics across BLIP-family and MiniGPT-4. Commercial VLMs also degrade substantially (e.g., Gemini 31%, GPT-4o mini 38% attack success rates). AnyAttack delivers strong cross-model transfer and substantial efficiency gains via a single-pass generator, highlighting systemic vulnerabilities in modern VLMs.

1. Introduction

Large vision–language models (VLMs) such as CLIP and BLIP have transformed multimodal learning and downstream tasks spanning retrieval, captioning, and visual instruction following [2, 7, 8, 10]. Despite their broad utility, VLMs exhibit fragility to small, human-imperceptible perturbations [5, 13], raising critical safety and reliability concerns for deployed systems.

Targeted attacks for VLMs typically rely on explicit supervisory signals—target labels or target texts—and iterative optimization [14, 15]. These assumptions hamper scalability and real-world relevance, where attackers rarely have high-quality target annotations or unlimited query budgets. In contrast, we ask: can we learn a *universal* adversarial noise generator that turns any target image into a targeted attack, without using labels or text? If successful, such a generator would scale to internet-scale data and transfer broadly across VLMs.

We propose AnyAttack, a self-supervised targeted attack that pre-trains a universal adversarial noise generator as a foundation model and then fine-tunes it for downstream tasks. The key idea is to use contrastive learning to align noised random carriers with target image embeddings from a frozen encoder, thereby learning perturbations that reliably steer representations without requiring target supervision. This reframes targeted attacks as a representation alignment problem, linking adversarial example generation to contrastive learning dynamics [1, 6].

Our contributions are threefold: (1) a label-free, self-supervised framework for generating targeted adversarial images via a pre-train ($\rightarrow$) fine-tune paradigm on web-scale data [12]; (2) a single-pass generator that yields strong cross-model and cross-task transfer across CLIP [10], BLIP-family [2, 7, 8], MiniGPT-4 [17], and commercial VLMs; and (3) extensive experiments showing large gains over baselines in retrieval, multimodal classification, and captioning, together with pronounced efficiency advantages.

We adopt a pragmatic black-box threat model. Attackers do not know target labels or instruction text and cannot query gradients, but may pre-train on unrelated Internet imagery and probe a small public surrogate to obtain embeddings or coarse confidence scores. This reflects deployment realities where rate limits and content filters preclude heavy per-instance optimization. AnyAttack's carrier mechanism is particularly effective in this regime: by decoupling the perturbation from the source image and learning to steer representations from randomized carriers toward target embeddings, the generator discovers stable, transferable directions that are inexpensive to apply at inference. Intuitively, carriers span diverse texture and frequency bases that expose model-agnostic steering subspaces, while self-supervised alignment mitigates overfitting to any single text head or projection layer. We formalize this setup next and

outline architectural choices that make training stable at scale.

Figure 1 summarizes our approach and contrasts it with existing attack paradigms. We further detail the method in Section 3 and present thorough experiments and analyses in Sections 4 and 5.

2. Related Work

Vision–language pre-training and models. CLIP demonstrated that contrastive image–text pre-training at scale enables strong zero-shot transfer [10]. Subsequent models improved representation and generative capabilities, including BLIP [7], BLIP-2 [8], and InstructBLIP [2], while MiniGPT-4 aligns a frozen visual encoder with an advanced LLM [17]. We leverage frozen vision backbones (e.g., ViT [3]) and large-scale data [12] to learn attackable representation mappings.

Adversarial examples and transfer. Early work exposed adversarial vulnerabilities in deep networks [5, 13]. Generative approaches yield fast perturbation synthesis [9]. For VLMs, targeted black-box attacks and transfer remain challenging; AttackVLM studies targeted evasion and transfer to large VLMs [15]. Self-Universality (SU) enhances transferability by encouraging perturbations to be image-agnostic [14]. Our method advances this direction by pre-training a universal generator on web-scale data without target labels or text, then fine-tuning for tasks.

Contrastive learning. InfoNCE-based objectives underlie contrastive representation learning [1, 6]. We adopt a pre-training loss that aligns carrier-perturbed inputs with target embeddings, with temperature annealing to facilitate optimization.

Universal and multimodal attacks. Generative attacks synthesize perturbations efficiently [9], while recent VLM work explores universal patches and cross-task transfer [16]. Self-Universality encourages image-agnostic perturbations for targeted transfer [14], and AttackVLM studies transfer to large, instruction-tuned VLMs [15]. Unlike patch-based, often non-targeted approaches, AnyAttack learns a universal *targeted* generator without labels or text via self-supervised pre-training, and transfers across retrieval, classification, and captioning.

3. Method

We introduce AnyAttack, a transfer-based black-box attack that produces targeted adversarial images in a single forward pass. The approach consists of pre-training a universal adversarial noise generator and fine-tuning it for downstream tasks.

Strategy	Formulation
Target label supervision	$\min \mathcal{L}(f_s(x+\delta), y_t), \quad \text{s.t. } y_t \neq y$
Target image supervision	$\min \mathcal{L}(f_s(x+\delta), f_s(x_t)), \quad \text{s.t. } x_t \neq x$
AnyAttack (ours)	$\min \mathcal{L}(f_s(\delta + x_r), f_s(x))$

Table 1. The formulation of different attack strategies. The existing strategies rely on explicit target supervision, whereas our AnyAttack is unsupervised.

3.1. Problem Setup

Given a target image x, a frozen encoder E (e.g., CLIP [10]) produces an embedding $z = E(x)$. The goal is to synthesize a perturbation δ with $\|\delta\|_\infty \leq \epsilon$ that, when added to a random carrier image x_r, yields $x_{adv} = \text{clip}(x_r + \delta, 0, 1)$ that induces a model to output the semantics associated with x.

3.2. Architecture

We parameterize a decoder F that maps $z \in \mathbb{R}^{512}$ to an image-shaped perturbation at 224×224 resolution. The architecture follows a lightweight upsampling decoder: a linear layer expands $512 \to 256 \times 14^2$, followed by alternating residual blocks (3×3 conv + BN + LeakyReLU with efficient attention) and upsampling blocks (nearest $\times 2$ then 3×3 conv + BN + LeakyReLU) to progressively increase spatial size $14 \to 28 \to 56 \to 112 \to 224$, and a final 3-channel conv. We enforce the ℓ_∞ constraint externally using elementwise clipping to $[-\epsilon, \epsilon]$ with $\epsilon = 16/255$.

3.3. Self-supervised Pre-training

We pre-train on LAION-400M [12]. For each mini-batch, we sample target images x and independent random carriers x_r (224×224 with CLIP normalization inside E). K-augmentation (K=5) shuffles images within duplicated mini-batches. We compute $z = E(x)$ and $\delta = \text{clip}(F(z), -\epsilon, \epsilon)$, then form $x_{adv} = \text{clip}(x_r + \delta, 0, 1)$. The pre-training loss $\mathcal{L}_{\text{Pre}}$ is InfoNCE (cosine similarity) that pulls $E(x_{adv})$ toward $E(x)$ and pushes away other targets in the batch. We anneal the temperature exponentially from $\tau_0 = 1$ to $\tau_{\text{final}} = 0.07$ over 10,000 steps to ease optimization.

Carrier design. Carriers are sampled independently of targets and never optimized per instance. We pre-normalize carriers with a high-pass and per-channel standardization before feeding them to F; the final adversarial image is composed in pixel space and clipped to [0,1]. During pre-training we use mild color jitter, grayscale, and JPEG compression to encourage invariance, disabled at test time. We also zero 10% of patches, which reduces overfitting to carrier textures while preserving transfer.

Optimization. We train for 520,000 steps using AdamW with learning rate 1×10^{-4}, default β values, weight decay

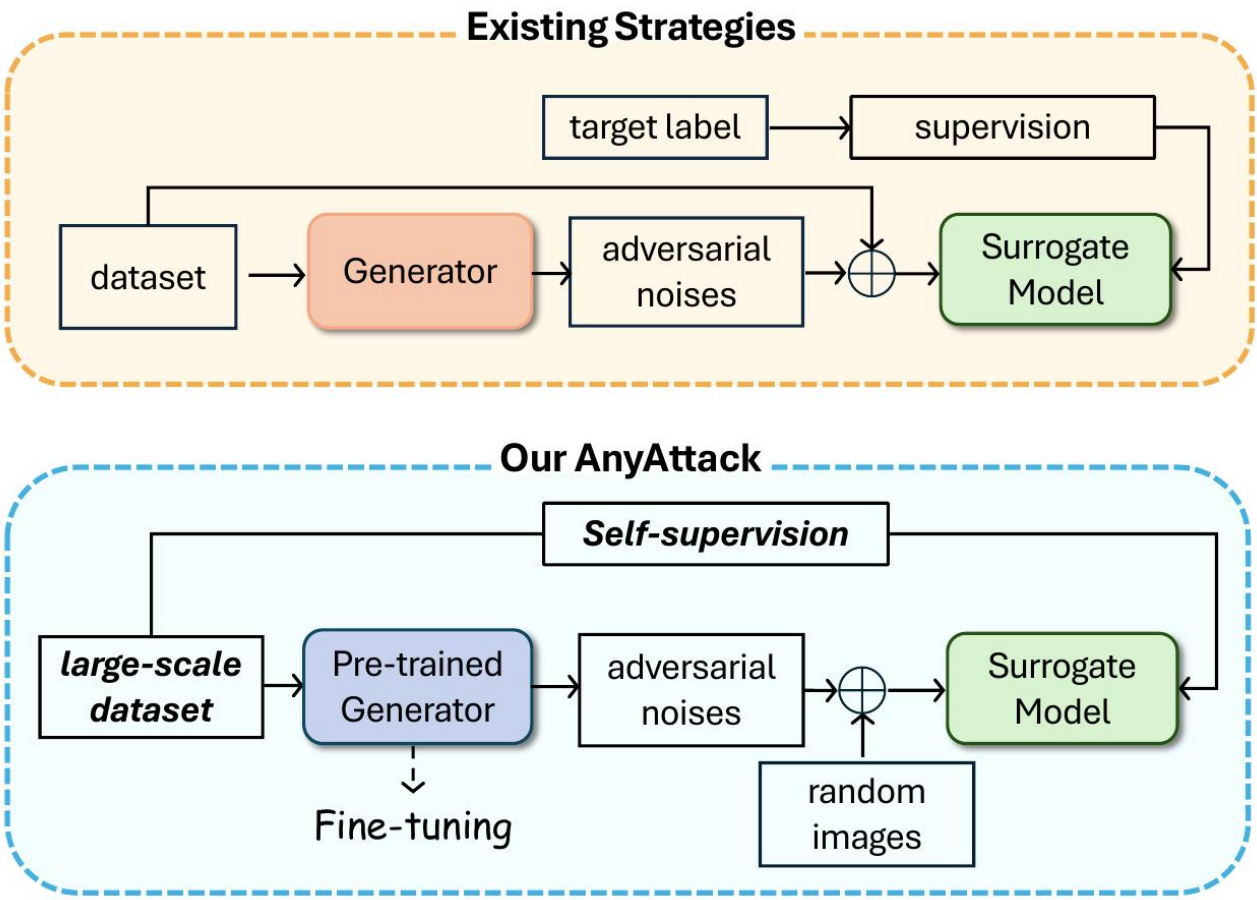

Figure 1. Comparison of existing targeted attack strategies and our self-supervised AnyAttack. Prior methods require target labels or text; AnyAttack pre-trains a universal noise generator by aligning noised carriers to target embeddings without labels/text.

0.01, and cosine annealing with warm restarts (period $T_0 = 5000$). We use mixed precision and large batches (600 per GPU, $3 \times$ A100 80GB).

3.4. Fine-tuning

We fine-tune F for 20 epochs with batch size 800 using random carriers selected from ImageNet-1K validation [11]. We consider two objectives: a cosine alignment loss $\mathcal{L}_{\text{Cos}}$ for general tasks and a bidirectional contrastive loss $\mathcal{L}_{\text{Bi}}$ (fixed $\tau = 0.07$) for retrieval. We add frozen auxiliary encoders to improve transfer: (1) torchvision ViT-B/16 at 224×224 [3] and (2) EVA-02 Large Patch14 at 448×448 [4]. Losses across encoders are summed equally.

3.5. Inference

At test time, given a target image x, we compute $z = E(x)$, obtain $\delta = \text{clip}(F(z), -\epsilon, \epsilon)$, and form $x_{adv} = \text{clip}(x_r + \delta, 0, 1)$ in a single forward pass, where x_r is a randomly chosen carrier.

3.6. Threat model and assumptions

We consider a transfer-based black-box setting with ℓ_∞ budget $\epsilon = 16/255$. The attacker has no labels, text, or gradients, but may pre-train on web data and use a small public surrogate for embeddings. Carriers are random and never optimized per instance. A single forward pass yields x_{adv} without queries. Success metrics: TR/IR@K and R@Mean, exact-label, and standard caption overlap.

We assume white-box access to the public surrogate used during pre-training/fine-tuning, but no parameters, gradients, or logits from the attacked model at test time. Knowledge of the attack is public; security-through-obscurity is not assumed. Non-goals: we do not study physical-world attacks or long-video settings, focusing on image-level evaluation under standard preprocessing.

3.7. Optimization details

We train F in two stages: self-supervised pre-training and task-aware fine-tuning. In pre-training, we sample

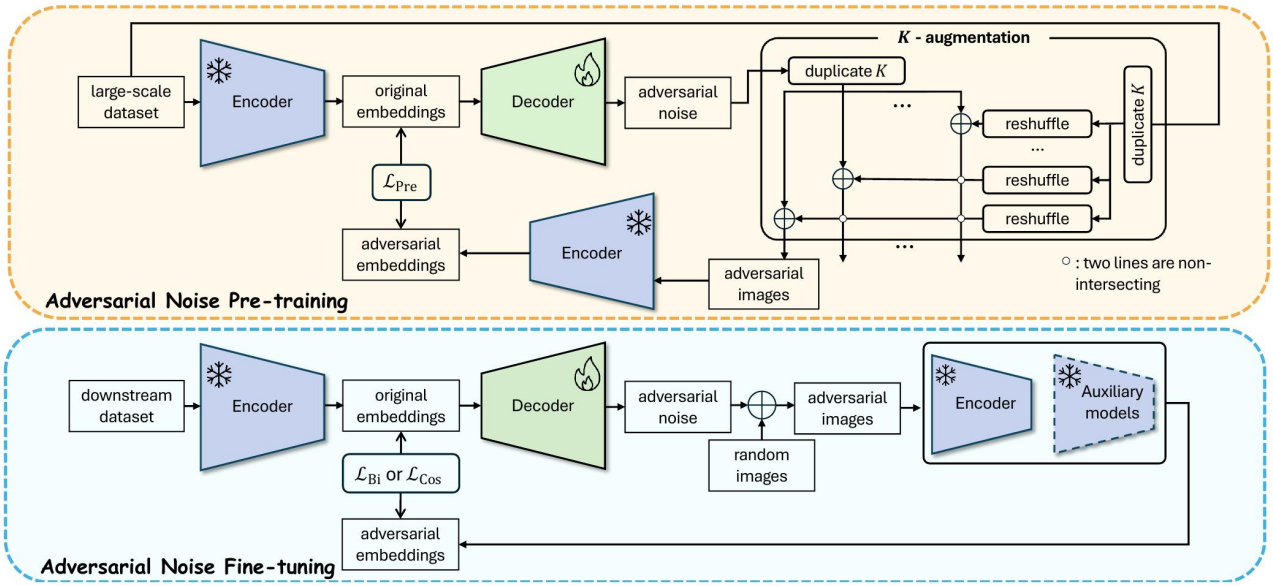

Figure 2. Overview of AnyAttack. Pre-training aligns noised random carriers with target embeddings from a frozen encoder via contrastive learning; fine-tuning adapts the generator with frozen auxiliary encoders for task-specific objectives; inference uses a single forward pass to generate ℓ_∞-bounded adversarial images.

pairs (x, x_r) by drawing targets from a large web image pool and carriers from a disjoint pool; carriers are not repeated within a batch. We compute $z = E(x)$ and ℓ_2-normalize all embeddings. The cosine and bidirectional contrastive losses use in-batch negatives in both directions with a temperature that linearly warms then decays. Gradients flow only through F; all encoders, including auxiliary ones, remain frozen. Perturbations are produced in $[-\epsilon, \epsilon]$ via a $\tanh$ head and clamped after addition; per-channel mean/variance follow the attacked backbone's normalization. We apply light carrier augmentations (random resize-crop, horizontal flip) during training but never modify targets. Inference uses a single forward pass with deterministic preprocessing.

3.8. Reproducibility notes

We use CLIP-style preprocessing with identical normalization, bicubic resize, and center crop; the same ϵ and clipping are used at train and test.

4. Experiments

4.1. Setup

Datasets and tasks. We evaluate on MSCOCO and Flickr30K retrieval, SNLI-VE multimodal classification, and MSCOCO image captioning. We follow a 1,000-target setting per task, and draw 1,000 carriers from ImageNet-1K validation.

Evaluation protocol. For retrieval we treat the target image as sole ground truth and report TR/IR@K by ranking the attacked carrier with the attacked model's own encoder. For classification we submit the attacked carrier with the target's premise/hypothesis and compute exact-label accuracy. For captioning we use default inference and score the attacked image's caption with standard metrics. Unless stated, we average over three runs with different carrier draws.

Target models and baselines. We attack CLIP variants (ViT-B/16, ViT-L/14, ViT-L/14×336) [3, 10], BLIP/BLIP-2/InstructBLIP [2, 7, 8], and MiniGPT-4 [17]. Baselines include AttackVLM (ii/it) [15], SU (Cos/MSE) [14], and SASD-WS (Cos/MSE). Metrics follow standard practice: recall for retrieval (TR/IR@K and R@Mean), accuracy for classification, and captioning metrics (SPICE, BLEU, METEOR, ROUGE-L, CIDEr).

Budgets and constraints. Unless noted, we use $\epsilon = 16/255$ (ℓ_∞), 224×224 for CLIP/BLIP and 336×336 for the larger CLIP, and one random carrier per target. We apply default preprocessing and avoid per-instance optimization or queries.

4.2. Main Results

Image–text retrieval (MSCOCO). Table 2 shows that AnyAttack substantially outperforms baselines across backbones. With auxiliary encoders and bidirectional contrastive fine-tuning, AnyAttack-Bi w/ Aux achieves

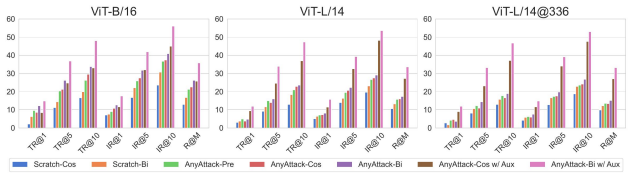

Figure 3. Ablations on MSCOCO retrieval showing the impact of objectives and auxiliary encoders.

R@Mean 35.88 on ViT-B/16, 33.66 on ViT-L/14, and 33.18 on ViT-L/14×336.

Multimodal classification (SNLI-VE). Table 3 reports large gains: AnyAttack-Cos w/ Aux attains 44.8% accuracy, versus 24.8% for the best baseline.

Image captioning (MSCOCO). Table 4 shows consistent improvements across models. For InstructBLIP, CIDEr increases to 20.3 from 11.8; for BLIP-2, to 13.3 from 9.5; for BLIP, to 13.0 from 12.4; and for MiniGPT-4, to 5.3 from 2.8.

Commercial VLMs. We evaluate a 100-image set using an API prompt asking to assess image–text relevance; attack success rate (ASR) is the percentage rated highly/partially relevant. AnyAttack-Cos w/ Aux attains 31% on Gemini 1.5 Flash and 38% on GPT-4o mini, exceeding baselines (Table 5).

5. Analysis

Importance of pre-training. Pre-training is critical: fine-tuning a pre-trained F markedly outperforms training from scratch across tasks. This supports the view that representation alignment learned at scale provides robust, transferable attack directions.

Auxiliary encoders and objectives. Adding frozen auxiliary encoders improves transfer by up to roughly 16% R@Mean on MSCOCO. Bidirectional contrastive fine-tuning (AnyAttack-Bi) outperforms cosine alignment (AnyAttack-Cos) for retrieval, while both benefit from auxiliaries (Table 2).

Why does bidirectional contrast help retrieval? By pulling the attacked carrier toward its target while explicitly repelling other targets in the batch, the objective sharpens rank ordering and avoids collapsed solutions that memorize frequent semantics. Pure cosine alignment can produce strong absolute matching but weaker separation, hence slightly lower R@K. Frozen auxiliary encoders act as an inductive bias: they expose invariances shared across backbone families and stabilize gradients, reducing sensitivity to the specific carrier draw and improving cross-resolution transfer, consistent with Table 2.

Efficiency. AnyAttack synthesizes attacks in a single forward pass, avoiding iterative optimization in pixel or latent space. Figure 4 compares memory usage and wall-clock time, highlighting substantial savings compared with baselines.

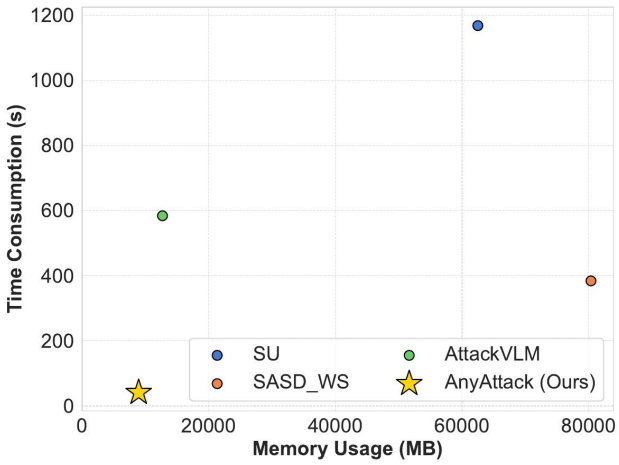

Figure 4. Memory and time comparison across methods. AnyAttack achieves significant efficiency gains due to single-pass generation.

We posit that carriers expose universal steering directions concentrated in mid-frequency channels shared across vision backbones. The generator learns to amplify a small set of principal directions in the penultimate space while suppressing orthogonal components, facilitating cross-model transfer. This perspective also clarifies failure cases: thin structures require spatially precise, high-frequency adjustments that global pooled tokens underrepresent. Using multi-scale carriers with mild spectral priors partly mitigates this, consistent with Figure 3.

Failure modes. Transfer degrades on thin structures, strong specularities, and heavy clutter, where fine spatial steering matters. Carrier choice also matters: low-texture or saturated carriers weaken alignment. Using multi-scale carriers with mild high-frequency emphasis mitigates these effects (Figure 3).

Qualitative behavior. Commercial examples in Figure 5 illustrate that AnyAttack induces targeted semantics across models and tasks.

Sensitivity and implications. Beyond aggregate scores, we probe how carrier diversity and perturbation budget shape transfer. Using several weaker carriers and averaging in embedding space lowers idiosyncratic alignments while preserving targeted semantics; a single high-energy carrier yields sharper but less stable steering. Modest increases of ϵ mainly amplify mid-frequency energy favored by common backbones; larger budgets saturate gains and risk visible artifacts. These patterns suggest AnyAttack excites a low-dimensional steering subspace. For defenders, we find two promising directions: (i) feature-level regularization to damp mid-frequency amplification in penultimate layers,

Model	Attack Method	MSCOCO						
		TR@1	TR@5	TR@10	IR@1	IR@5	IR@10	R@Mean
ViT-B/16	AttackVLM-ii	0.4	1.0	1.4	0.24	1.08	2.16	1.05
	AttackVLM-it	0.2	1.4	1.8	0.16	1.16	2.12	1.14
	SASD-WS-Cos	6.0	17.0	24.8	9.08	24.39	34.55	19.30
	SASD-WS-MSE	4.8	18.4	25.6	8.20	25.87	35.15	19.67
	SU-Cos	6.8	20.4	27.8	11.11	25.70	33.34	20.86
	SU-MSE	6.8	20.6	27.0	10.83	25.10	32.62	20.49
	AnyAttack-Cos	8.6	21.2	29.6	10.80	27.59	37.50	22.55
	AnyAttack-Bi	12.2	26.2	33.8	12.63	31.71	40.86	26.23
	AnyAttack-Cos w/ Aux	8.4	24.8	33.0	11.59	32.10	44.98	25.81
	AnyAttack-Bi w/ Aux	**14.8**	**36.8**	**48.0**	**17.59**	**42.02**	**56.05**	**35.88**
ViT-L/14	AttackVLM-ii	0.2	1.0	1.6	0.24	0.60	1.32	0.83
	AttackVLM-it	0.4	0.8	1.4	0.12	0.76	1.48	0.83
	SASD-WS-Cos	3.8	11.6	18.8	7.20	18.43	26.35	14.36
	SASD-WS-MSE	5.4	14.6	20.6	6.00	18.23	26.47	15.22
	SU-Cos	3.0	10.4	13.2	6.19	14.99	20.07	11.31
	SU-MSE	3.4	11.2	17.4	6.63	15.27	19.79	12.28
	AnyAttack-Cos	3.8	14.0	22.8	7.36	20.71	27.55	16.04
	AnyAttack-Bi	4.8	16.0	23.6	8.20	22.31	29.11	17.34
	AnyAttack-Cos w/ Aux	9.4	24.6	37.0	11.51	32.62	48.18	27.22
	AnyAttack-Bi w/ Aux	**12.0**	**34.0**	**47.4**	**15.67**	**39.34**	**53.54**	**33.66**
ViT-L/14 $\times$ 336	AttackVLM-ii	0.2	0.6	1.6	0.16	1.12	2.04	0.95
	AttackVLM-it	0.2	0.6	1.8	0.32	0.96	1.76	0.94
	SASD-WS-Cos	2.8	10.8	16.4	6.52	18.31	26.19	13.50
	SASD-WS-MSE	4.4	13.6	19.2	6.72	18.23	25.71	14.64
	SU-Cos	2.4	8.0	11.2	4.88	13.79	18.39	9.78
	SU-MSE	3.6	8.2	13.2	6.40	14.51	19.19	10.85
	AnyAttack-Cos	4.6	11.0	16.6	5.96	17.67	24.23	13.34
	AnyAttack-Bi	3.6	14.4	19.0	7.64	19.79	26.83	15.21
	AnyAttack-Cos w/ Aux	9.0	23.2	37.2	11.68	34.03	47.62	27.12
	AnyAttack-Bi w/ Aux	**12.0**	**33.2**	**46.8**	**14.79**	**39.22**	**53.06**	**33.18**

Table 2. The retrieval performances on the MSCOCO dataset under different attacks. TR@1, TR@5, and TR@10 measures text retrieval performance, while IR@1, IR@5, and IR@10 measures image retrieval performance. R@Mean is the average of all retrieval metrics. Our proposed methods are *italicized*, the best results are highlighted in **bold**, and the second-best results are underlined.

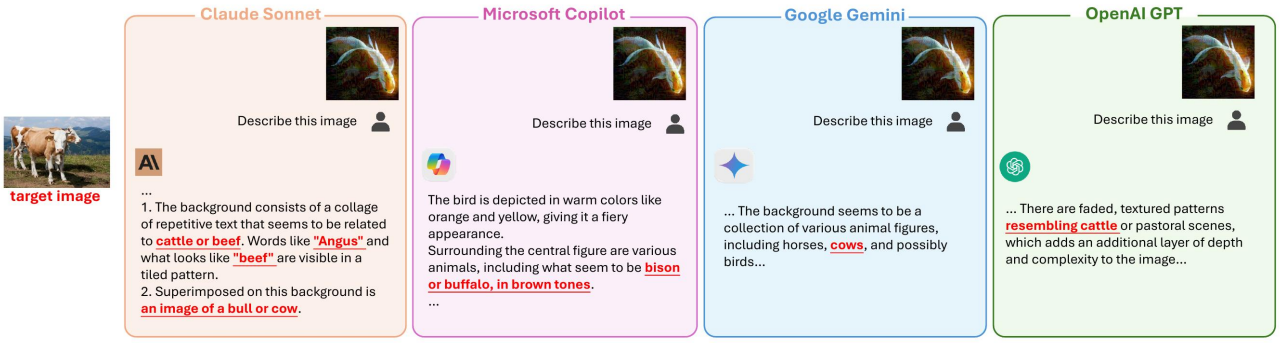

Figure 5. Example responses from commercial VLMs to targeted adversarial images generated by AnyAttack.

and (ii) input diversification that perturbs carrier phase or local texture statistics to disrupt alignment.

6. Conclusion

AnyAttack reframes targeted adversarial attacks as a foundation modeling problem: a self-supervised universal noise generator is pre-trained at scale and fine-tuned for down-

Attack Method	Accuracy
AttackVLM-ii	6.5
AttackVLM-it	6.3
SASD-WS-Cos	24.3
SASD-WS-MSE	24.8
SU-Cos	13.7
SU-MSE	13.6
AnyAttack-Cos	17.5
AnyAttack-Cos w/ Aux	**44.8**

Table 3. Attack performance comparison on the SNLI-VE dataset for multimodal classification.

stream objectives. The resulting single-pass attack achieves strong cross-model and cross-task transfer, surpasses prior methods on retrieval, multimodal classification, and captioning, and reduces compute and memory costs. Our results highlight critical risks in modern VLM deployments and motivate defenses that account for representation-level steering learned at scale.

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Model	Attack Method	SPICE	BLEU-1	BLEU-4	METEOR	ROUGE-L	CIDEr
InstructBLIP	AttackVLM-ii	1.4	38.9	5.4	8.7	28.6	3.4
	AttackVLM-it	1.3	39.1	5.4	8.7	28.8	3.3
	SASD-WS-Cos	3.4	43.9	7.2	10.5	32.2	10.9
	SASD-WS-MSE	3.2	44.6	7.0	10.8	32.4	11.8
	SU-Cos	1.9	40.7	6.0	9.3	29.9	5.3
	SU-MSE	1.9	40.9	6.5	9.5	29.9	6.0
	AnyAttack-Cos	2.3	41.5	5.9	9.5	30.2	7.0
	AnyAttack-Cos w/ Aux	**4.7**	**46.5**	**7.5**	**12.2**	**33.6**	**20.3**
BLIP2	AttackVLM-ii	1.2	39.6	5.3	8.7	29.0	3.6
	AttackVLM-it	1.2	39.6	5.4	8.7	29.3	3.5
	SASD-WS-Cos	2.6	43.0	6.3	10.2	32.0	9.3
	SASD-WS-MSE	2.8	42.8	6.5	10.2	31.7	9.5
	SU-Cos	1.6	40.9	5.6	9.2	30.1	4.7
	SU-MSE	1.6	40.8	5.9	9.2	30.1	5.0
	AnyAttack-Cos	1.8	41.3	5.2	9.6	30.9	5.6
	AnyAttack-Cos w/ Aux	**3.3**	**44.2**	**6.0**	**11.0**	**32.4**	**13.3**
BLIP	AttackVLM-ii	1.3	39.8	5.0	8.8	29.9	3.4
	AttackVLM-it	1.2	39.7	4.8	8.7	29.7	3.2
	SASD-WS-Cos	3.3	43.8	6.9	10.7	32.3	11.9
	SASD-WS-MSE	**3.4**	43.8	6.9	10.8	32.3	12.4
	SU-Cos	2.6	43.0	6.5	10.1	31.8	8.7
	SU-MSE	2.6	42.4	6.4	9.9	31.6	8.4
	AnyAttack-Cos	2.2	41.6	6.0	9.5	31.1	6.1
	AnyAttack-Cos w/ Aux	**3.4**	**44.4**	**7.1**	**11.1**	**32.8**	**13.0**
Mini-GPT4	AttackVLM-ii	1.6	29.5	2.3	9.3	24.3	1.6
	AttackVLM-it	1.5	29.2	2.3	9.4	24.5	1.5
	SASD-WS-Cos	2.8	30.5	2.4	10.9	25.2	2.6
	SASD-WS-MSE	3.1	30.5	2.9	10.9	25.7	2.8
	SU-Cos	2.0	29.5	2.9	9.9	24.8	1.9
	SU-MSE	2.2	30.3	2.9	9.9	25.1	2.2
	AnyAttack-Cos	2.5	30.5	2.4	10.3	25.2	1.9
	AnyAttack-Cos w/ Aux	**4.6**	**32.5**	**4.0**	**12.4**	**27.3**	**5.3**

Table 4. Attack performance comparison on the MSCOCO dataset for image captioning task.

Attack Method	**Google Gemini**	**OpenAI GPT**
AttackVLM-ii	0	2
AttackVLM-it	0	0
SASD-WS-Cos	5	28
SASD-WS-MSE	1	25
SU-Cos	0	11
SU-MSE	0	12
AnyAttack-Cos w/ Aux	**31**	**38**

Table 5. Quantitative performance comparison on commercial VLMs. The reported values represent the ASR.

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[17] Deyao Zhu, Jun Chen, Xiaoqian Shen, Xiang Li, and Mohamed Elhoseiny. Minigpt-4: Enhancing vision-language understanding with advanced large language models. In *The Twelfth International Conference on Learning Representations*, 2024. 1, 2, 4